

LLM-Based Windows Log Analysis

This project explored using Large Language Models (LLMs) to detect suspicious activity in Windows Security logs and compare the results with Wazuh. Wazuh was used as a general baseline because it uses a traditional rule-based alerting system, while the LLMs reviewed logs and assigned suspicion scores on a scale from 0 to 15.

The goal of the project was to detect suspicious activity in Windows logs, identify the top 5 most suspicious events, generate human-readable summaries explaining why logs were suspicious, and compare LLM results with Wazuh alerts.

Technologies Used

Python, Pandas, Wazuh, OpenAI GPT-4 Turbo, Anthropic Claude, Windows Event Logs, Matplotlib, and Virtual Machines were used throughout the project.

Features

The project processed over 10,000 Windows Security log entries and generated suspicion scores on a 0–15 scale. The LLMs created summaries explaining suspicious activity, identified the top 5 most suspicious events, and compared the results against Wazuh alerts. Charts and graphs were also created to visualize the score distributions between the datasets and detection systems.

Project Overview

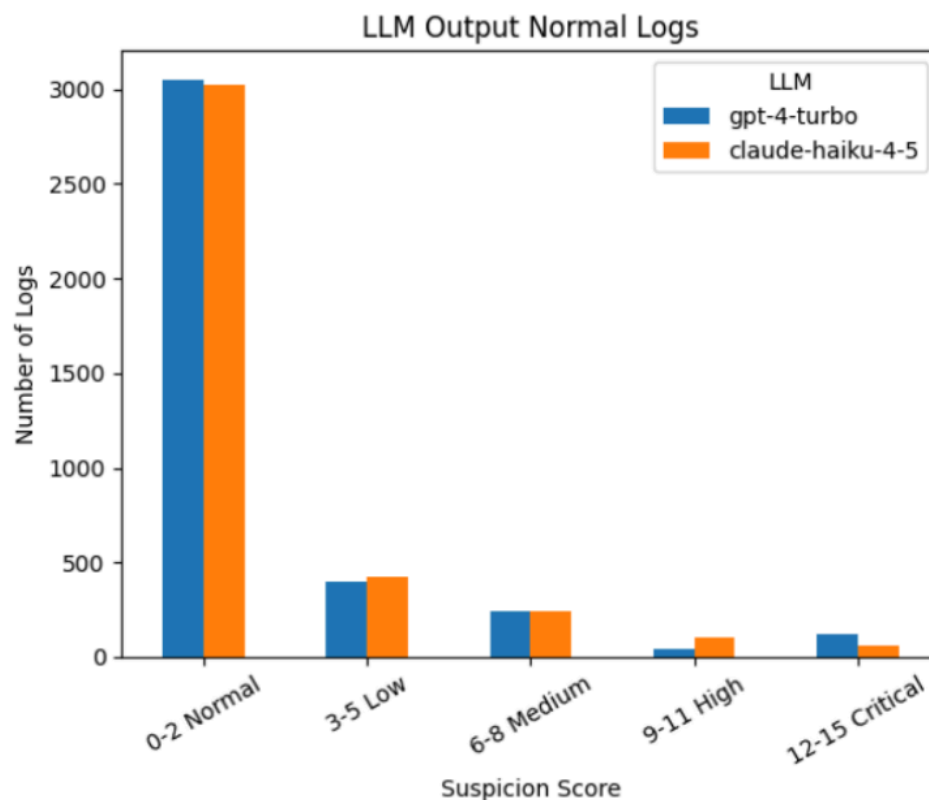
Windows Security logs were collected from virtual machines running both normal and suspicious activity. The logs were converted from EVTX format into CSV files and processed using Python and Pandas before being sent to the LLMs for analysis.

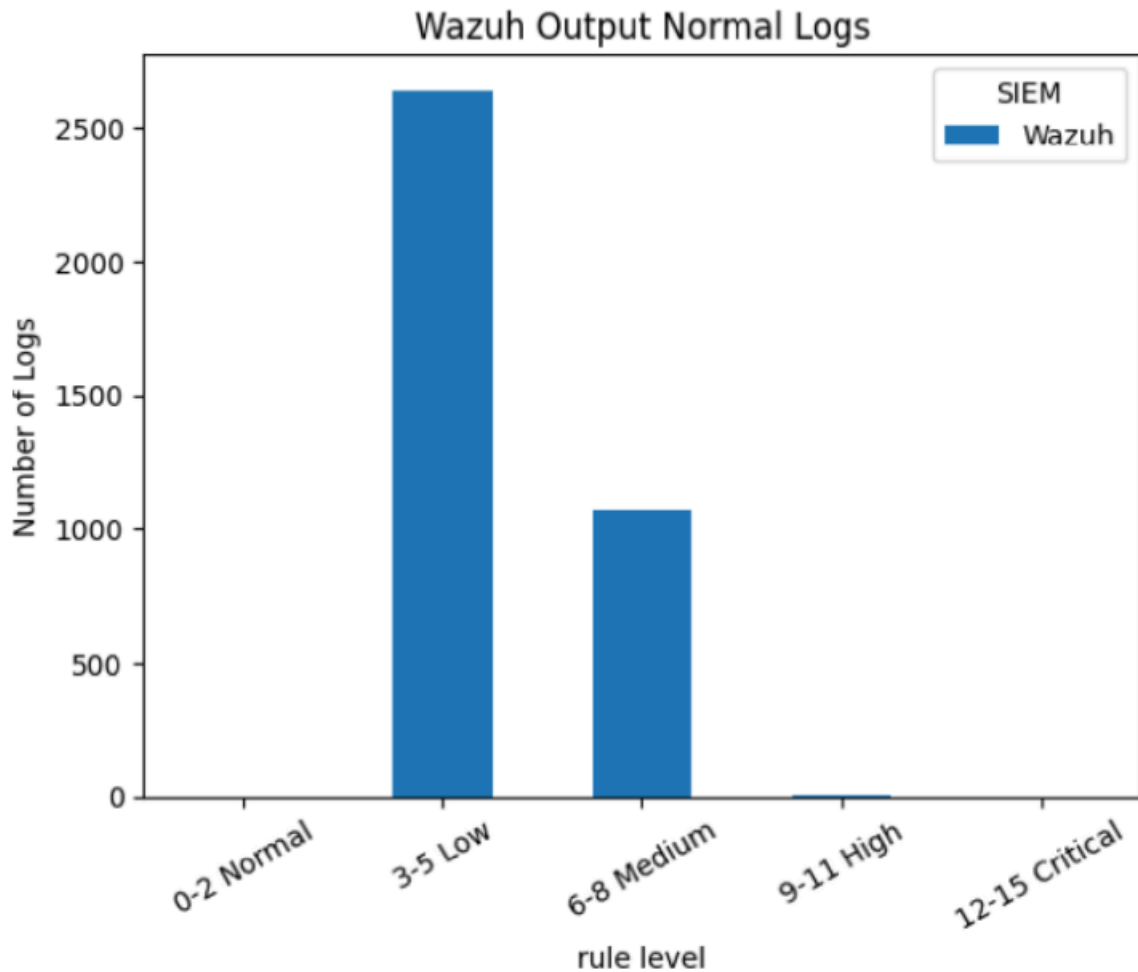
The suspicious dataset included failed login attempts, user account creation, administrator group changes, scheduled task persistence, PowerShell activity, and system reconnaissance commands.

The LLMs analyzed the logs and generated suspicion scores, human-readable summaries, and top 5 suspicious events. The results were then compared against Wazuh alerts to evaluate differences between LLM-based analysis and traditional rule-based detection systems.

Normal Dataset Results

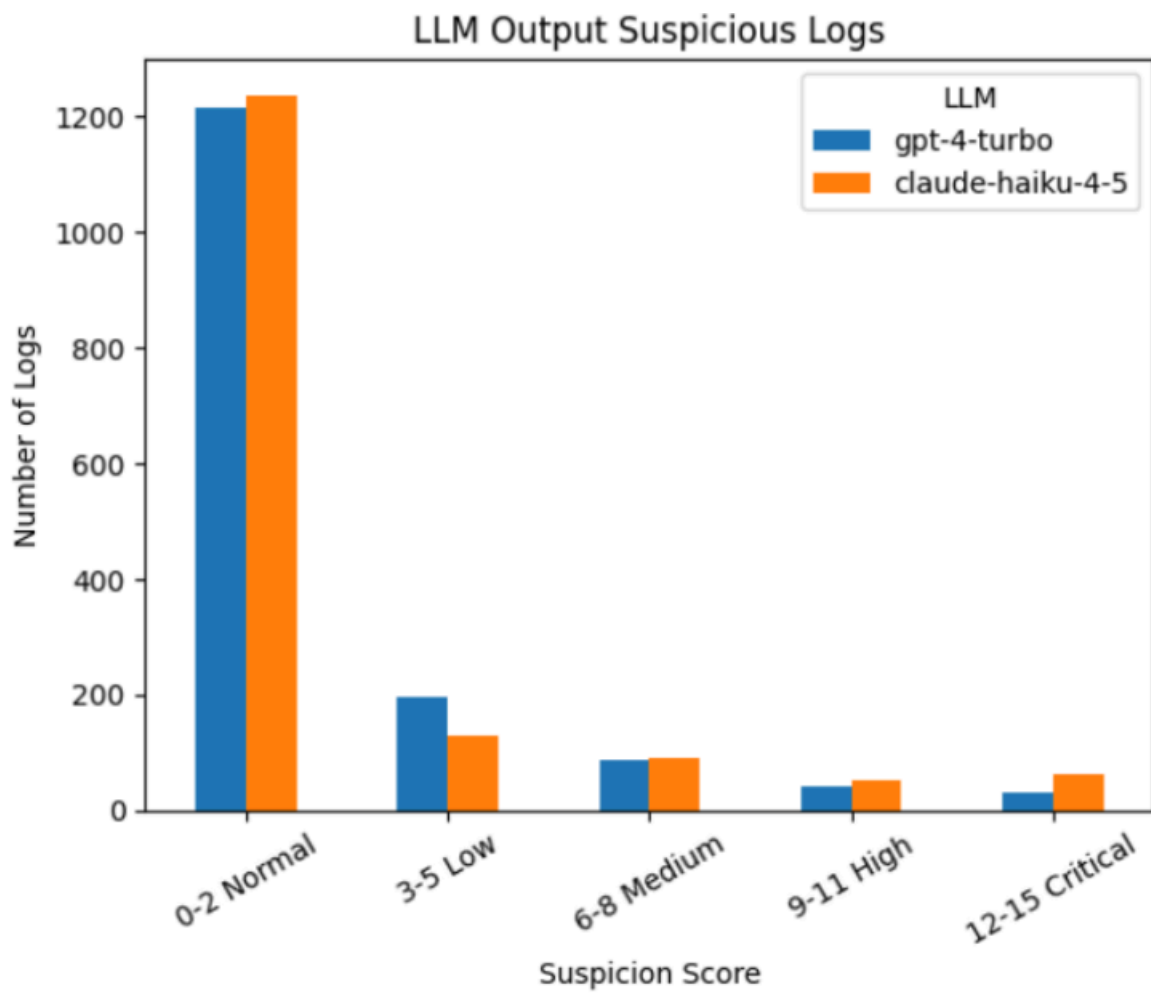
Overall, both LLMs showed similar distributions in how they scored the logs. However, both models also tended to score normal logs as more suspicious than they actually were. In some cases, the models assigned high suspicion scores even when there was no suspicious activity in the logs. Wazuh mostly generated low and medium level alerts for the normal dataset.

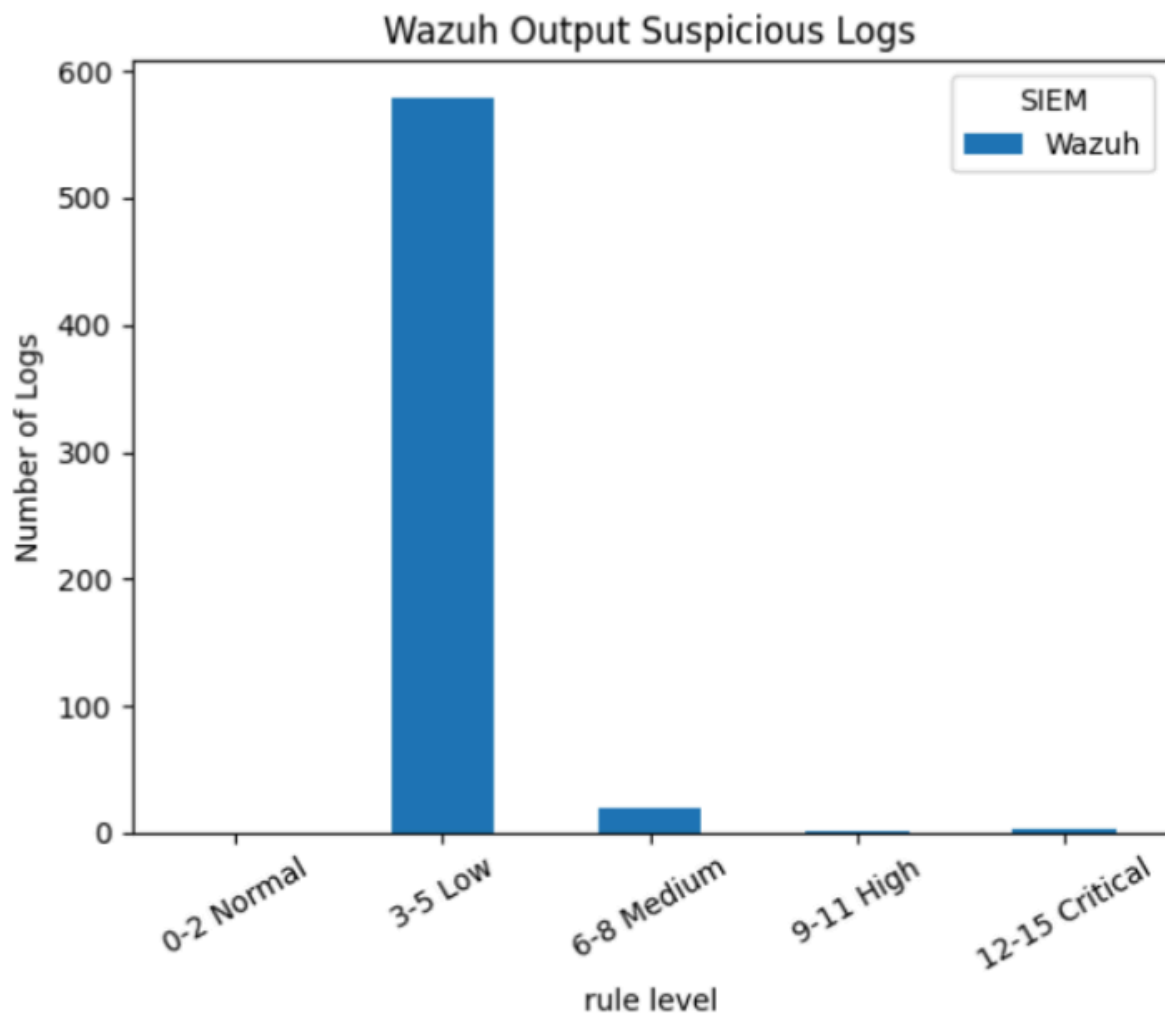




Suspicious Dataset Results

For the suspicious dataset, both LLMs successfully detected suspicious activity and generated useful summaries explaining why the logs were flagged. Some false positives and incorrect detections were also observed during testing. Wazuh generated several alerts for failed logins, administrator group changes, and suspicious system activity. The suspicious dataset produced noticeably higher suspicion scores compared to the normal dataset.





Top 5 Results

Normal Dataset

Claude

RowID	Suspicion_summary					
194	13	svchost using suspicious email credential pizzahuthacker@gmail.com				
663	11	MSI installer from PowerShell temp directory				
742	13	wazuh-agent spawning powershell querying password policy				
83	7	Upfc.exe with suspicious launchtype boot parameter				
679	8	ICACLS reset permissions ossec-agent directory				

GPT

RowID	Suspicion	summary
194	15	Suspicious email in username
597	15	User added to Administrators
742	10	Security policy export and deletion
708	9	Critical file permission removal
83	12	Unusual executable path

Wazuh

rule\level	rule\id	rule\description
11	521	Possible kernel level rootkit
9	19005	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (27)
9	19005	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (27)
9	19011	CIS Benchmark for Amazon Linux 2023 Benchmark v1.0.0.: Ensure message of the day is configured
9	19005	SCA summary: CIS Microsoft Windows 11 Enterprise Benchmark v3.0.0: Score less than 30% (27)

For GPT, all events that were in the top 5 did match the logs. However, the suspicious score for the logs was much higher than it should have been. The summaries for the logs also match closely with what was expected.

For Claude, there were several hallucinations of events that had not occurred. There was no user added to administrators, and no critical file permissions were removed. The summaries from Claude were noticeably less detailed compared to GPT.

Wazuh had a very low level alert for these logs. The VMs were unhardened, and there were several benchmark warnings for being out of compliance. There was a possible kernel-level rootkit warning, but this is due to the VMs not being hardened.

Overall, for the normal dataset, the LLMs were saying normal activity was very suspicious and hallucinating events that never happened. One other thing of note was that the email for the Windows account was pizzahuthacker, and that was being flagged as a highly suspicious event. The account, however, only did normal activity. The LLMs were overemphasizing the name of the account and not looking at the context of what it was doing. For the top 5, the prompt was changed to have less emphasis on the name of emails; otherwise, the entire top 5 would have been the email's username.

Suspicious dataset
Claude

RowID	Suspicion	summary			
1371	15	Adding global group to administrators via net			
1372	15	Net1 adding global to administrators group			
1377	15	Adding system2 user to administrators group			
1367	14	Scheduled task creation for persistence via cmd			
1447	14	PowerShell creating scheduled task Updater			

GPT

RowID	Suspicion	summary			
718	12	Suspicious user login attempt			
1367	12	Suspicious scheduled task creation			
1227	12	Suspicious executable in SoftwareDistribution			
318	12	Suspicious PowerShell command			
1361	12	Suspicious conhost forced execution			

Wazuh

rule\level	rule\id	rule\description			
12	60154	Administrators Group Changed			
12	60154	Administrators Group Changed			
12	60154	Administrators Group Changed			
10	60204	Multiple Windows Logon Failures			
9	19011	CIS Benchmark for Amazon Linux 2023 Benchmark v1.0.0			

For Claude, it successfully detected the suspicious activity and did not have any hallucinations in the top 5. Additionally, the summaries did match closely with what was done in the attack.

For GPT, it successfully detected the suspicious activity, but it did have hallucinations in the top 5. There was no suspicious executable installed on the host. The summaries did match the suspicious activity in the logs.

Wazuh had several alerts for changes in the administrators' group and multiple Windows login failures. Wazuh detected the suspicious activity as expected.

Overall, both LLMs detected suspicious activity and created useful summaries for why it was suspicious. Claude did say the suspicious logs were much more suspicious compared to GPT.

Cost Analysis

The project also evaluated the estimated cost of using LLMs for large-scale log analysis. Running full log analysis directly through LLMs was extremely expensive, especially when processing all logs instead of filtering suspicious activity first.

GPT-4 Turbo cost approximately \$8.77 to analyze the normal dataset and \$4.19 for the suspicious dataset. Claude Haiku cost approximately \$3.37 for the normal dataset and \$0.82 for the suspicious dataset.

In real-world environments, LLMs would more likely be paired with a SIEM system so that only suspicious activity is sent to the model for summarization and analysis.